Unified Modeling Language (UML)

Faculty of Computer & Information
Software Engineering
Third year

1- What is UML

- The Unified Modeling Language (UML) is a graphical language for visualizing, specifying, constructing, and documenting the artifacts of a software-intensive system.

- The UML offers a standard way to write a system's blueprints, including
  - **Conceptual** things such as business processes and system functions
  - **Concrete** things such as programming language statements, database schemas .

2- Goals of UML

- Provide users with a ready-to-use, expressive visual modeling language so they can develop and exchange meaningful models.
- Be independent of particular programming languages and development processes.
- Provide a formal basis for understanding the modeling language.

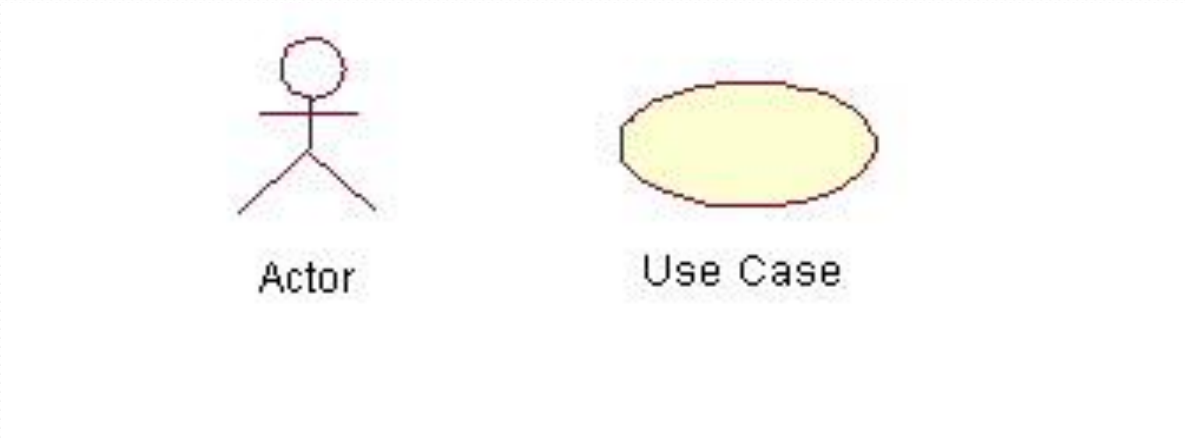
3-UML Diagrams

1) Use Case Diagram .
2) Activity Diagram .
3) Class Diagram .
4) Communication Diagram .
5) Component Diagram .
6) Deployment Diagram .
7) Object Diagram .
8) Package Diagram .
9) Sequence Diagram .
10) State Diagram .

4-Use Case Diagram

- A use case is a set of scenarios that describing an interaction between a **user** and a **system**. It displays the relationship among **actors** and **use cases**.

- The two main components of a use case diagram are **use cases** and **actors**



4-Use Case Diagram

- **Use Case**

  A use case describes a sequence of actions that provide something of measurable value to an actor .



- **Actor**

  An actor is a person, organization, or external system that plays a role in one or more interactions with your system



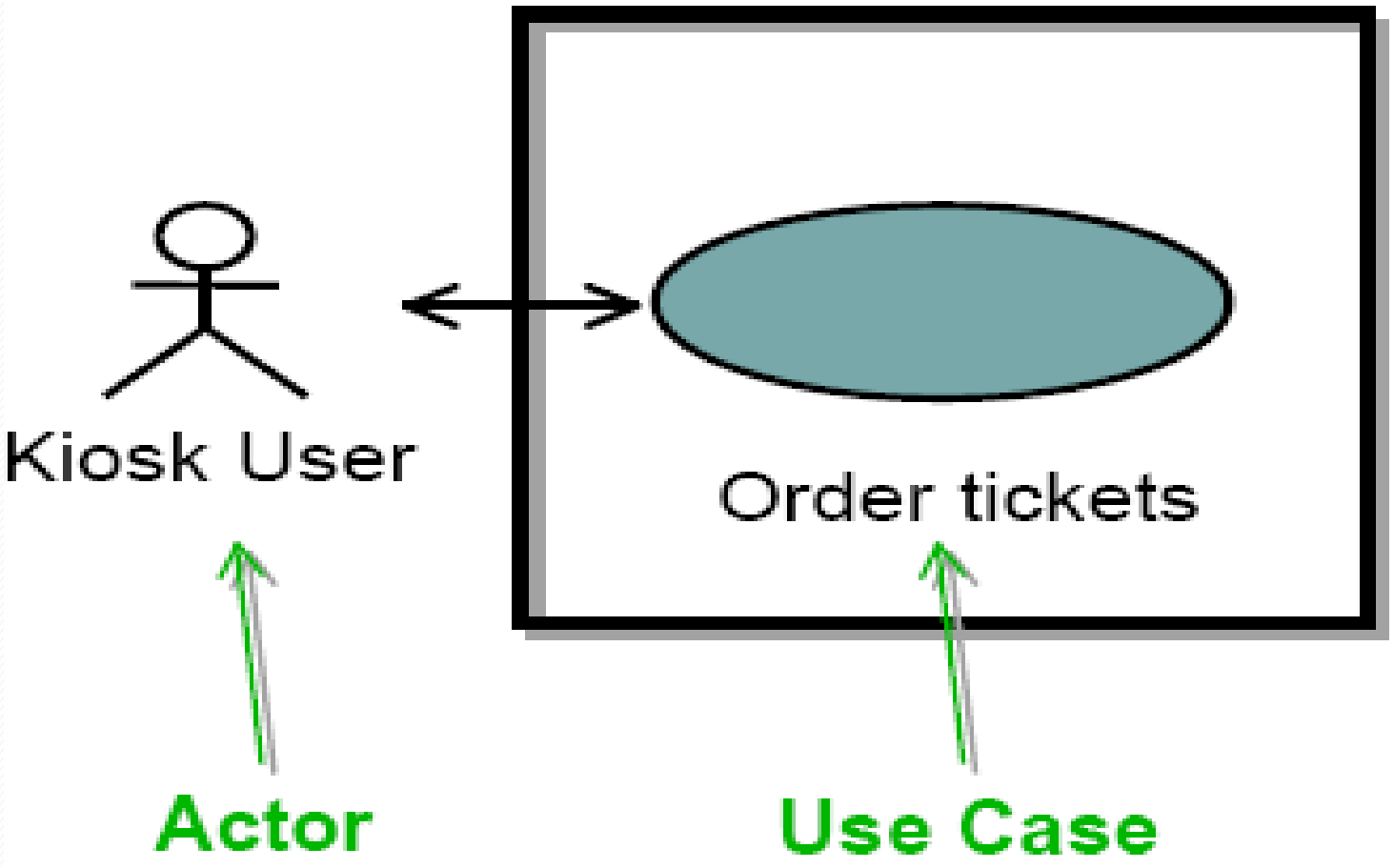
5-Identifying Use Cases

- Which functions does the actor require from system?
- Does the actor need to read, create, destroy, modify, or store some kind of information in the system?
- Does the actor have to be notified about events in the system

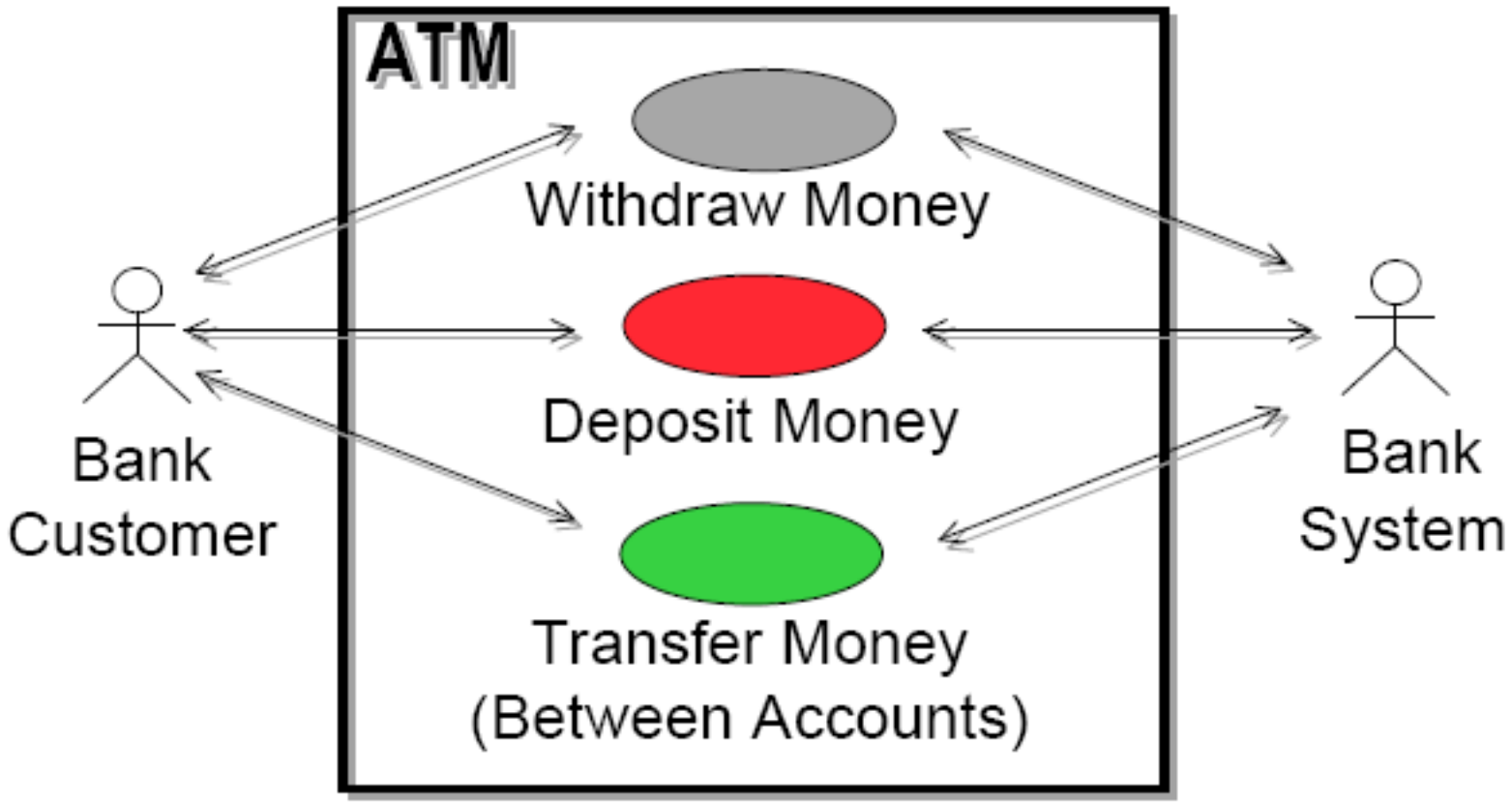
6-Identifying actors

- Useful questions
  - **Who will use the main functionality of the system(primary actors)?**
  - **Who will need to maintain, administrate, and keep the system working (secondary actors)?**
  - **Which hardware devices does the system need to handle?**
  - **With which other systems does the system need to interact?**
  - **Who or what has an interest in the results (the value) that the system produces?**

7-Use Case Examples



8-Use Case Examples

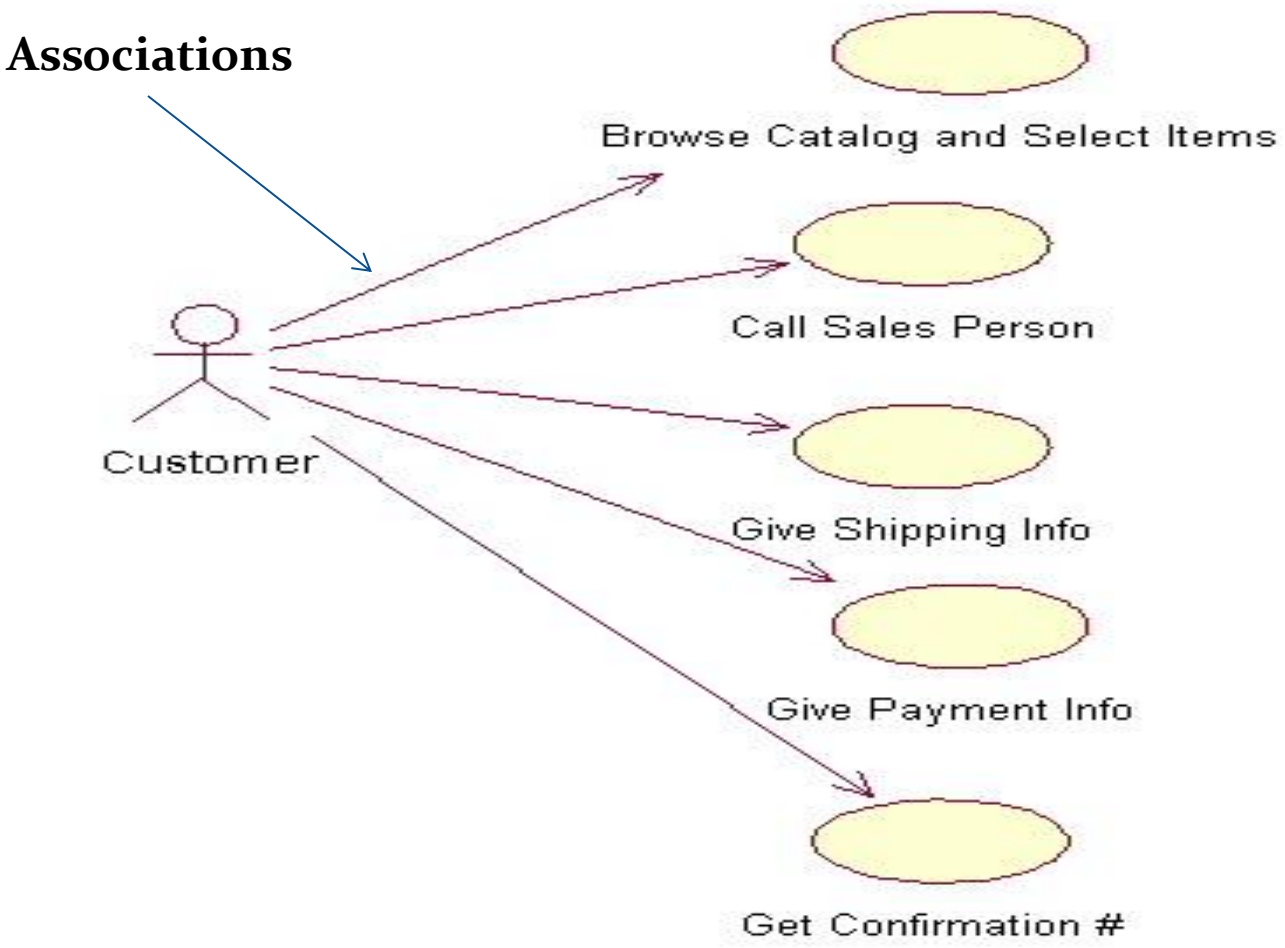


9- Creating Use Case Diagrams

- Start by listing a sequence of steps a user might take in order to complete an action. For example a user placing an order with a sales company might follow these steps.

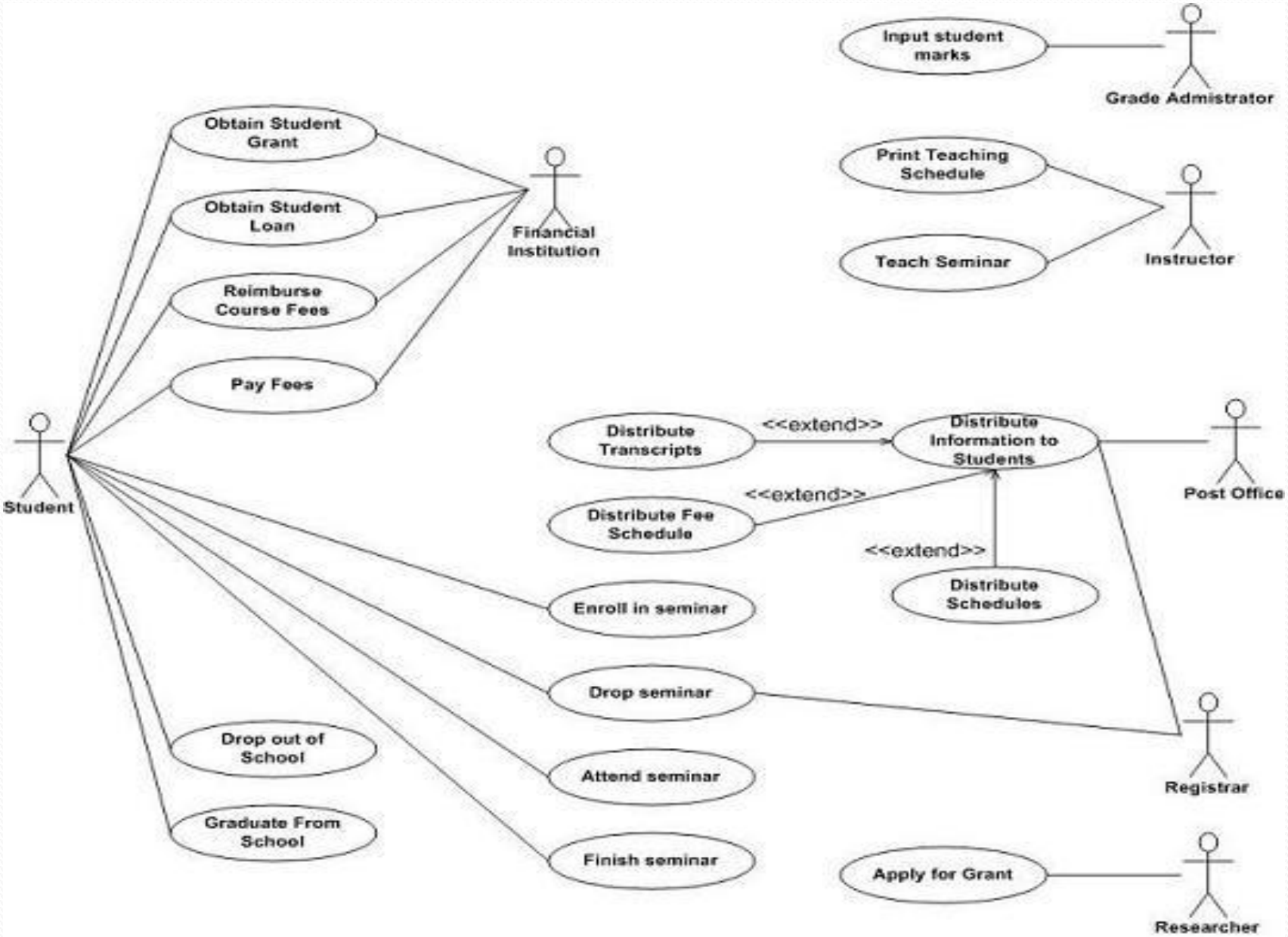
1. **Browse catalog and select items.**
2. **Call sales representative.**
3. **Supply shipping information.**
4. **Supply payment information.**
5. **Receive conformation number from salesperson.**

9- Creating Use Case Diagrams



10- Advanced Example

- Students are enrolling in courses with the potential help of registrars.
- Professors input the marks students earn on assignments
- registrars authorize the distribution of transcripts (report cards) to students


Input student marks
Grade Admistrator
Obtain Student Grant
Obtain Student Loan
Reimburse Course Fees
Pay Fees
Financial Institution
Print Teaching Schedule
Teach Seminar
Instructor
Student
Distribute Transcripts
<<extend>>
Distribute Information to Students
Post Office
Distribute Fee Schedule
<<extend>>
<<extend>>
Distribute Schedules
Enroll in seminar
Drop seminar
Drop out of School
Attend seminar
Graduate From School
Finish seminar
Registrar
Apply for Grant
Researcher

Bank System

- A bank has several automated teller machines (ATMs), which are geographically distributed and connected via a wide area network to a central server.

- Each ATM machine has a
  - **Card reader**
  - **Cash dispenser**
  - **keyboard/display**
  - **Receipt printer.**

- By using the ATM machine, a customer can
  - **Withdraw cash from either a checking or savings account**
  - **Query the balance of an account**
  - **Transfer funds from one account to another.**

Bank System

- A transaction is initiated when a customer inserts an ATM card into the card reader.
- Encoded on the magnetic strip on the back of the ATM card are the **card number**, the **start date**, and the **expiration date.**
- Assuming the card is recognized, the system validates the ATM card to determine that
  - **The expiration date has not passed,**
  - **The user-entered PIN (personal identification number) matches the PIN maintained by the system**
  - **The card is not lost or stolen.**
- The customer is allowed three attempts to enter the correct PIN; the card is confiscated if the third attempt fails.
- Cards that have been reported lost or stolen are also confiscated.

Bank System

- If the PIN is validated satisfactorily, the customer is prompted for a withdrawal, query, or transfer transaction.

- Before a withdrawal transaction can be approved, the system, determines that
  - **Sufficient funds exist in the requested account,**
  - **The maximum daily limit will not be exceeded**
  - **There are sufficient funds at the local cash dispenser.**

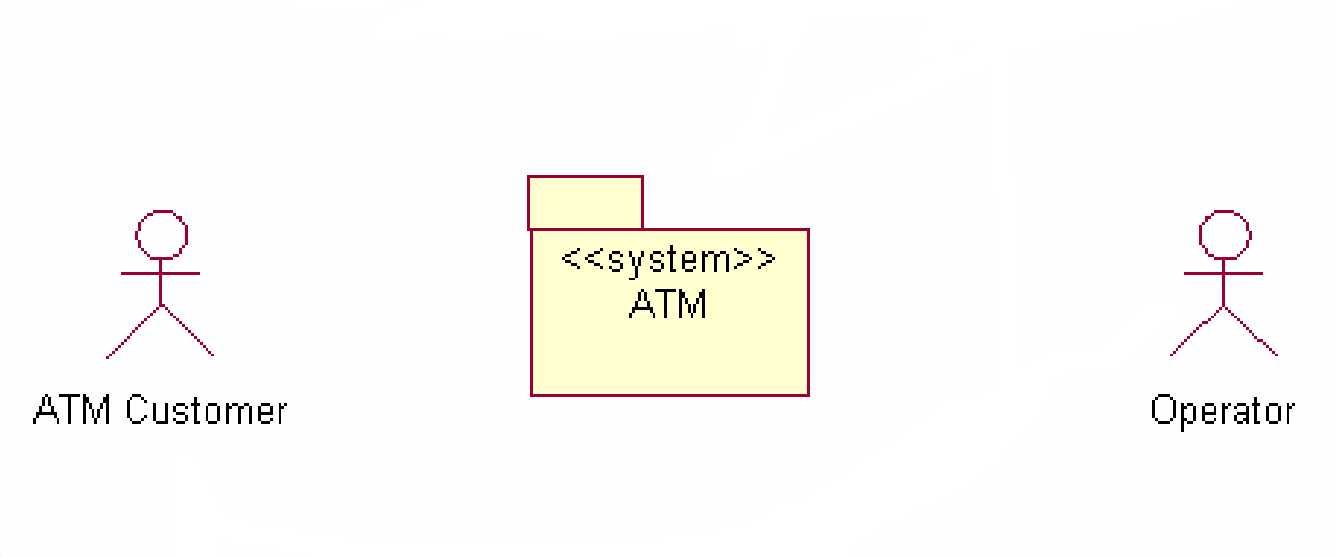
- If the transaction is approved, the requested amount of cash is dispensed, a receipt is printed containing information about the transaction, and the card is ejected.

- Before a transfer transaction can be approved, the system determines that the customer has at least two accounts and that there are sufficient funds in the account to be debited.

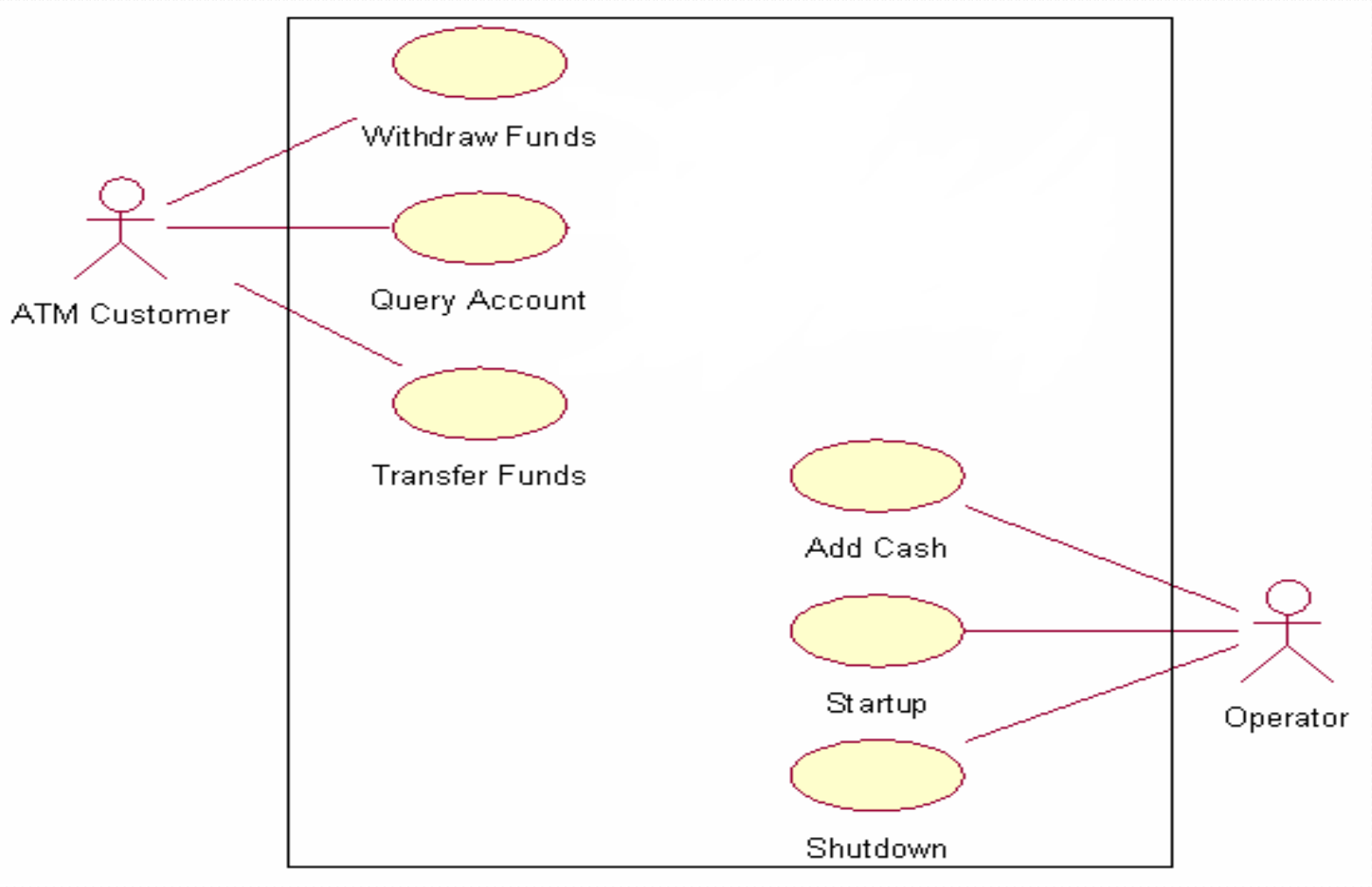
Bank System

- For approved query and transfer requests, a receipt is printed and the card is ejected.
- A customer many cancel a transaction at any time; the transaction is terminated and the card is ejected.
- Customer records, account records, and debit card records are all maintained at the server.
- An ATM operator may start up and close down the ATM to replenish the ATM cash dispenser and for routine maintenance.
- It is assumed that functionally to open and close accounts and to create, update, and delete customer and debit card records is provided by an existing system and is not part of this problem.

ATM System



ATM: Use Case Model



ATM: Use Case Model

